Customer Behavior Analysis for Alfido Tech

1. Introduction

Customer behavior analysis helps businesses understand how customers interact with their products and services. By analyzing transaction data, companies can identify purchasing patterns, segment customers into meaningful groups, and detect churn risks.

The objective of this project is to analyze customer transaction data and identify different customer segments based on their purchasing behavior. RFM analysis and K-Means clustering are used to group customers according to their purchase recency, frequency, and monetary value. Additionally, visualizations are used to study sales trends, product performance, payment preferences, and churn distribution.This analysis helps Alfido Tech improve customer engagement, design targeted marketing campaigns, and increase customer retention.

2. Dataset Description

The dataset used in this project contains customer transaction data collected from an e-commerce platform. The dataset includes multiple variables that describe customer purchasing behavior.

Some of the important attributes in the dataset include:

- Customer ID – Unique identifier for each customer
- Purchase Date – Date when the purchase was made
- Product Category – Category of product purchased
- Total Purchase Amount – Total amount spent by the customer
- Payment Method – Method used for payment
- Churn – Indicates whether a customer has stopped purchasing from the platform

The dataset provides useful information to analyze purchasing trends and customer behavior.

3. Data Cleaning and Preprocessing

Before performing analysis, the dataset was cleaned to ensure data quality and consistency. Duplicate records were removed to avoid repeated observations. Rows containing missing values were also removed to prevent errors during analysis.

Column names were standardized by removing unnecessary spaces. The column "Customer Age" was removed because it was not required for this particular analysis.The purchase date column was converted into datetime format to enable time-based analysis such as calculating recency values and identifying monthly sales trends.These preprocessing steps ensured that the dataset was ready for feature engineering and clustering.

4. Feature Engineering Using RFM Analysis

RFM analysis is a commonly used technique in marketing analytics to understand customer purchasing behavior.

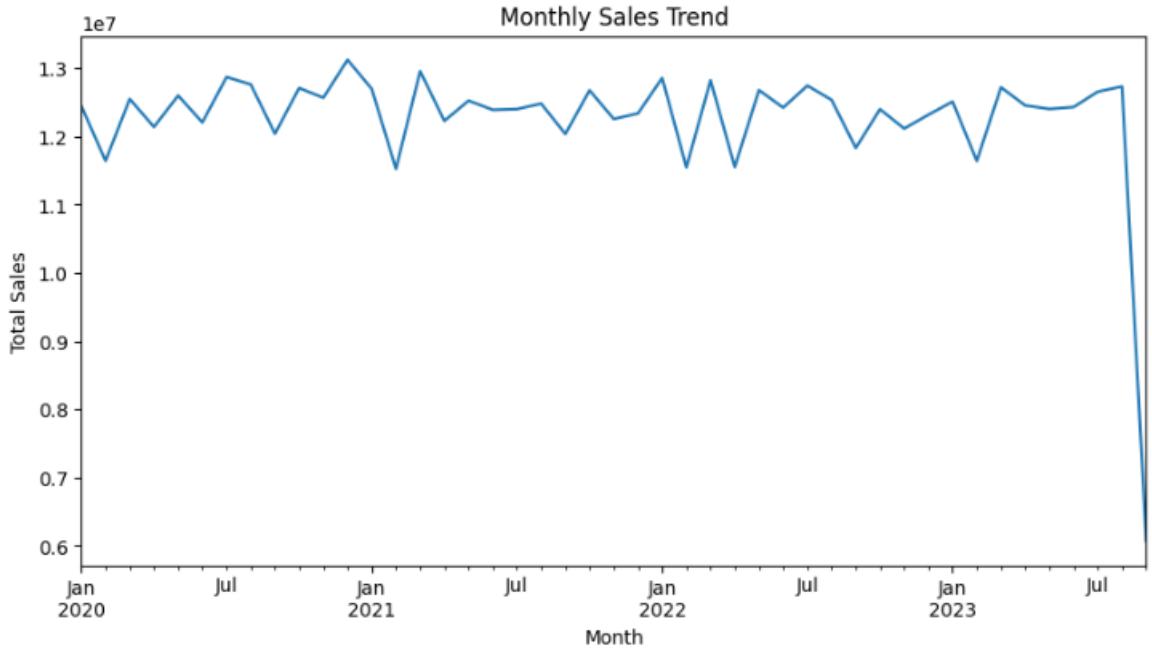
1. Recency measures how recently a customer made a purchase. Customers who purchased recently are more likely to make another purchase.
2. Frequency measures how often a customer makes purchases. Customers with higher purchase frequency tend to be more loyal.
3. Monetary value measures the total amount spent by a customer. Customers with higher spending contribute significantly to company revenue.

These three features were calculated for each customer and used as input for the clustering algorithm.

5. Customer Segmentation Using K-Means Clustering

K-Means clustering was used to segment customers based on their RFM values. Before applying the clustering algorithm, the RFM values were standardized using StandardScaler to ensure that all variables were on the same scale.

The Elbow Method was used to determine the optimal number of clusters. Based on the elbow graph, four clusters were selected to segment customers into different groups.Each cluster represents customers with similar purchasing behavior. These clusters were later



interpreted as Loyal Customers, Regular Customers, New Customers, and At Risk Customers.

6. Sales Trend Analysis

Sales trend analysis helps identify how revenue changes over time. Monthly sales were calculated by grouping the purchase data by month.

The visualization shows fluctuations in sales across different months. Such trends may indicate seasonal purchasing behavior or the impact of marketing campaigns.

7. Product Category Analysis

Product category analysis helps identify which product categories generate the highest revenue. By analyzing sales across categories, businesses can understand which products are most popular among customers.

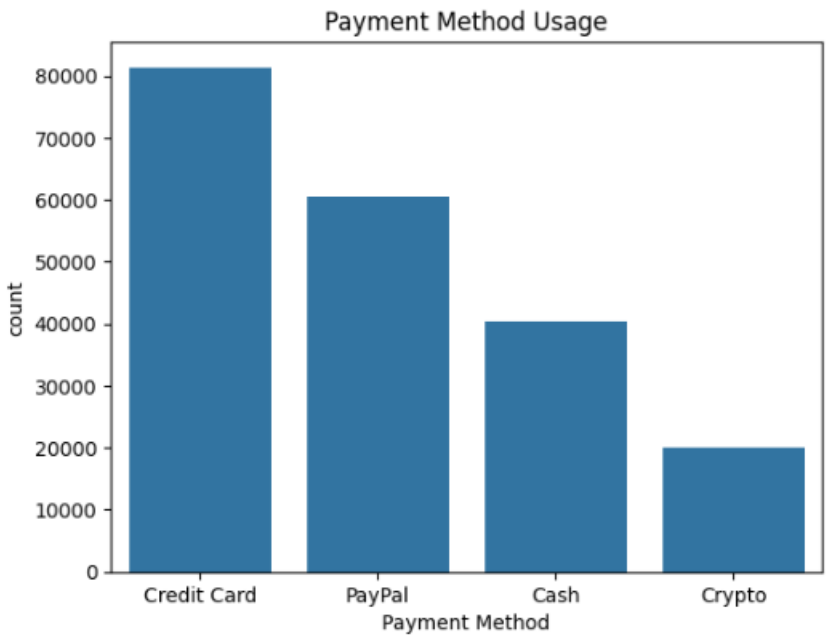
The visualization highlights the categories that contribute most to total sales.

High-performing categories may require additional marketing focus, while lower-performing categories may require strategic improvements.

8. Payment Method Analysis

Payment method analysis shows the distribution of payment options used by customers. Understanding payment preferences helps businesses improve the checkout process and ensure smooth transactions.

The analysis shows which payment methods are most frequently used by customers.

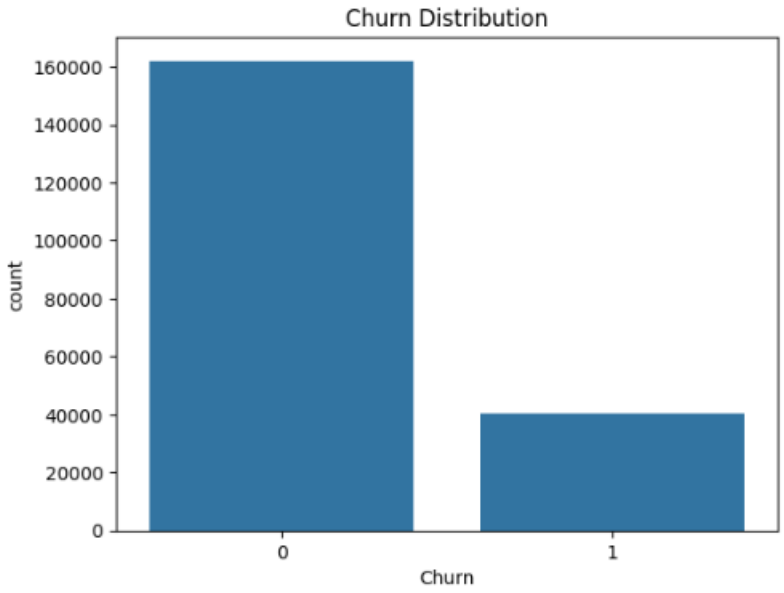


9. Churn Analysis

Churn analysis helps identify customers who have stopped purchasing from the platform. Understanding churn patterns is important because retaining existing customers is often more cost-effective than acquiring new ones.

The churn distribution visualization shows the number of customers who have churned compared to those who remain active.

This analysis helps identify potential retention strategies.



10. Key Insights

From the analysis, several insights were observed regarding customer behavior and purchasing patterns.

Customer segmentation revealed that customers can be grouped into different categories based on their purchase recency, frequency, and spending behavior. Some customers purchase frequently and contribute significant revenue, while others purchase less frequently or have stopped purchasing recently.

Monthly sales analysis showed variations in revenue across months, suggesting possible seasonal demand patterns. Product category analysis identified the most popular categories among customers. Payment method analysis indicated customer preference toward certain digital payment methods.

Churn analysis highlighted the presence of customers who may stop purchasing if not properly engaged.

11. Recommendations for Alfido Tech

Based on the findings from the analysis, several recommendations can be suggested to improve customer engagement and retention.

1. Alfido Tech can implement personalized marketing campaigns by targeting different customer segments with customized promotions. Loyal customers can be rewarded with loyalty programs or exclusive offers to maintain engagement.
2. At-risk customers can be re-engaged through reminder emails, discounts, or special promotions to encourage them to return.
3. Product recommendation systems can be implemented to suggest relevant products based on previous purchases, which can increase average order value.
4. Seasonal promotions can be planned based on sales trends to maximize revenue during peak purchasing periods.

Improving the checkout experience by optimizing commonly used payment methods can enhance customer satisfaction and reduce transaction friction.

12. Conclusion

This project analyzed customer transaction data to understand purchasing behavior and identify customer segments. Using RFM analysis and K-Means clustering, customers were grouped based on their purchasing patterns.

The analysis provided insights into sales trends, product performance, payment preferences, and churn behavior. These insights can help Alfido Tech design better marketing strategies, improve customer engagement, and increase retention.

Customer segmentation is an important tool for businesses because it allows companies to focus their efforts on the most valuable customers while also addressing potential churn risks.